LAWDECK

AI Cost Explainer

What Claude will cost Lawdeck, how it is charged, and what it means for pricing

Prepared: 20 October 2026 · **Companion file:** Lawdeck AI Cost Model.xlsx

Status: Planning estimate. Figures use stated assumptions and will be replaced by measured costs from the pilot.

1. Summary

Lawdeck uses Anthropic's Claude as its AI. Claude is paid for per use, in US dollars, like prepaid units: no use, no cost. The cost of one student message ranges from under ₦3 to about ₦190, depending on which Claude model handles it and what the student asked for.

> **Key finding**
>
> Which Claude model handles which task is the largest cost decision in the business. Using the top model (Opus) for everything costs a heavy user about ₦22,000 a month in AI alone; a lean mix costs about ₦4,400 for the same work.
>
> Only the lean mix supports a price students can realistically pay. The first build phase therefore tests whether the cheaper Claude models give good enough answers for each task.
>
> All costs are in US dollars while revenue will be in naira, so the exchange rate is a direct risk to margin.

All naira figures assume ***₦1,500 = US$1****, a placeholder. Change it in the spreadsheet's Inputs sheet.*

2. How Claude is charged

- **Pay as you go.** Lawdeck loads credit into an Anthropic account. Each AI request deducts a small amount. There is no monthly subscription for the AI itself.
- **Charged by "tokens".** A token is a piece of a word. 1,000 tokens is about 750 words. Prices are quoted per one million tokens (about 750,000 words).
- **Reading and writing are priced separately.** Input is everything Claude reads to answer (instructions, the question, library passages, recent chat). Output is what Claude writes back, including its hidden reasoning ("thinking"). Writing costs five times more than reading.
- **One message** means one student question plus one Lawdeck reply. A conversation of five questions is five messages.

Claude model	Input (US$ per 1M tokens)	Output (US$ per 1M tokens)	Intended use
Claude Opus 5.5	$4.00	$20.00	Marking answers; producing documents
Claude Sonnet 5.5	$2.00	$10.00	Everyday explanations, if quality holds

Claude model	Input (US$ per 1M tokens)	Output (US$ per 1M tokens)	Intended use
Claude Haiku 5.5	$0.10	$0.50	Small jobs; possibly explanations

Source: Anthropic API price list, October 2026.

3. One message, worked through

A student asks: “Explain the pre-action protocol in Lagos.”

Step	Calculation (Claude Opus 5.5)
Claude reads about 8,000 tokens (instructions 1,500; question 50; library passages 6,000; recent chat 500)	8,000 ÷ 1,000,000 × $4 = $0.032
Claude writes about 1,500 tokens of answer, plus about 30% more for its thinking (1,950 tokens)	1,950 ÷ 1,000,000 × $20 = $0.039
Total for this message	$0.071, about ₦107

Cost of one message by task and model

Task	Opus 5.5	Sonnet 5.5	Haiku 5.5
Everyday explanation	$0.071 (₦107)	$0.036 (₦53)	$0.0018 (₦3)
Marking a practice answer	$0.092 (₦138)	$0.046 (₦69)	$0.0023 (₦3)
Producing a document	$0.128 (₦192)	$0.064 (₦96)	$0.0032 (₦5)
Small jobs (routing, flashcards, titles)	$0.009 (₦14)	$0.005 (₦7)	$0.0002 (under ₦1)

4. Cost per student per month

Example heavy user: 200 messages a month (150 explanations, 30 marked answers, 10 documents, 10 small jobs).

Scenario	Models used	AI cost per month	Price needed*
A. Opus for everything	Opus for all tasks	$14.78 (₦22,173)	₦73,910
B. Mixed	Sonnet explains; Opus marks and drafts; Haiku small jobs	$9.37 (₦14,051)	₦46,837
C. Lean	Haiku explains and small jobs; Sonnet marks; Opus drafts documents	$2.93 (₦4,393)	₦14,643

** Monthly price at which AI is 30% of the price, the target set in the PRD.*

Lighter users cost proportionally less

Messages per month	A. Opus	B. Mixed	C. Lean
20	₦2,217	₦1,405	₦439

Messages per month	A. Opus	B. Mixed	C. Lean
60	₦6,652	₦4,215	₦1,318
200	₦22,173	₦14,051	₦4,393

5. Fixed running costs

Item	Per month
Vercel (app hosting)	about $20
Supabase (database, sign-in, file storage)	about $25
Voyage AI (library search)	about $5 allowance
Domain name	about $1.25 ($15 a year)
Total	about $51 (₦76,875)

These exist even with no students. Hosting prices are approximate and should be checked.

6. Example month (illustrative only)

Scenario C (Lean); 500 paying students at ₦5,000 a month, averaging 80 messages each; 1,000 free users limited to 15 messages each; 1.5% payment fee.

Line	Per month
Revenue	₦2,500,000
Payment processing fees	₦37,500
AI cost: paying students	₦878,565
AI cost: free users	₦329,462
Fixed running costs	₦76,875
Total costs	₦1,322,402
Profit	₦1,177,598 (47% margin)
AI cost as a share of revenue	48% (above the 30% target)
Paying students needed to break even	129

The example is profitable, but AI takes 48% of revenue, above the 30% target. That points to one or more of: a higher price, lower monthly message limits, a free tier restricted to the cheapest model, or a cheaper model mix. None of these figures are decisions; they show how the levers interact. Staff, marketing and legal costs are not included.

7. Levers that control cost

- **Model choice per task.** The biggest lever (section 4). Tested in the first build phase.
- **Usage limits per plan.** Monthly message caps keep the heaviest users within budget.
- **Free tier on the cheapest model.** Free users cost almost nothing if served by Haiku.

- **Prompt caching.** Claude charges about a tenth of the normal input price for instructions it has seen recently, which cuts the reading cost of repeated instructions.
- **Shorter library passages.** Sending fewer, better-chosen passages lowers input cost.
- **Spending caps.** A hard monthly cap and daily alert on the Anthropic account prevent runaway bills.

8. Risks for the finance review

- **Exchange rate.** Costs are in dollars, revenue in naira. A weaker naira raises costs directly. Consider pricing reviews each term, or a buffer in the price.
- **Paying in dollars.** Anthropic and the hosting providers bill by card in US dollars; a card that supports dollar payments is needed.
- **Usage uncertainty.** Message counts and sizes are estimates until the pilot. Lawdeck records the cost of every AI call, so real figures will be available after the first weeks.
- **Thinking overhead.** Claude's reasoning is billed as output; 30% is an estimate.
- **Price changes.** AI prices have generally fallen over time, but this is not guaranteed.

9. Questions for you

1. What price range is realistic for Law School students, and should there be a per-term or exam-season option?
2. What exchange-rate buffer should the price include?
3. What target margin should we plan for once staff and marketing are included?
4. Which payment provider and methods (card, bank transfer, USSD) should we use, and what are their real fees?
5. How should we fund dollar payments to Anthropic and the hosting providers?

10. How to use the spreadsheet

- Change only blue cells; yellow cells are the key assumptions.
- Inputs: exchange rate, thinking overhead, target AI share, model prices, task sizes and fixed costs.
- Cost per Answer: cost of one message for each task and model.
- Scenarios: three model mixes; the model names can be changed from a drop-down list.
- Monthly Plan: choose a scenario, number of students, price and free-tier limits to see revenue, costs, profit and break-even.